Mishkal Goliath — User Manual

November 5, 2026 · Lovy Elias

Goliath is an all-in-one mixing and mastering processor by Lovy Elias: clean-up EQ, resonance control, eight character EQs, nine compressors, a de-esser, tone shaping, tape and a mastering limiter, in a chain you can put in your own order. This manual covers the beta (V84) for Mac, VST3 and Audio Unit.

Quick start



The full Goliath plugin

You can be working in under a minute:

1. Run **Install Mishkal Goliath Beta.pkg**. If macOS says it can't be opened, go to System Settings › Privacy & Security, click **Open Anyway**, then run it again.
2. Restart your DAW. In Ableton: Settings › Plug-ins › Rescan.
3. Load **Mishkal Goliath Beta** on a track, a bus or the master.
4. Click the preset name in the top bar and pick a starting point from the folder that matches your source (for a lead vocal, try Vocals › Lead Vocal Modern; for a finished song, Mastering › Master

Balanced).

5. Click **auto** under the Input knob and play the loudest part for a few seconds, or set Input by hand so the meters peak around -12 to -6 dB. Then adjust by ear.

Auto input. auto listens for a few seconds and sets Input so the strip is hit at about -18 dBFS on average with peaks under -6 dBFS, the level the presets are made for. The button shows what it set (for example *set +4.5*); click it again while it's listening to stop.



Input with the auto button

Requirements: macOS 12 or newer, Apple Silicon or Intel, a VST3 or Audio Unit host.

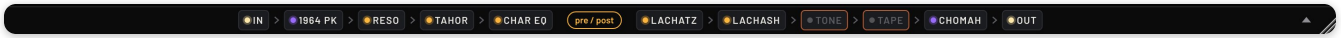
How Goliath is laid out

The top bar holds presets and comparing; the black display shows the EQ, the levels and the limiter; below it sit the seven modules, laid out in the order the sound passes through them by default. The strip along the bottom shows that order and lets you change it.

Module	What it's for
1 · Tahor EQ · Reso	Clean-up first: an 8-band surgical EQ, a resonance suppressor, the input level and the 1964 PK peak catcher.
2 · Character EQs	Colour and tone from eight hardware-style EQs.
3 · Lachatz (dynamics)	Nine compressors (four channel, five mastering and bus) plus the vocal tools: gate, breath and rider.
4 · Lachash (de-esser)	Sibilance and harsh peaks.

5 · Tone	Perach (adaptive tone) or Zerem (style shaper), in the drawer.
6 · Tape	Slil (tape machine) or Esh (character saturation), in the drawer.
7 · Chomah (limiter · output)	Goliath Classic, Multi or XL, true peak, clipper, loudness target, width and output.

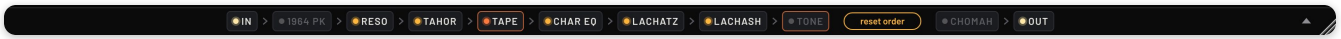
Building your chain



The signal-flow strip along the bottom of the plugin

The strip along the bottom is the signal path, left to right: IN › 1964 PK › RESO › TAHOR › CHAR EQ › LACHATZ › LACHASH › TONE › TAPE › CHOMAH › OUT. Lit chips are switched on; a stage that's off passes the sound through untouched. Click TONE or TAPE to open the drawer.

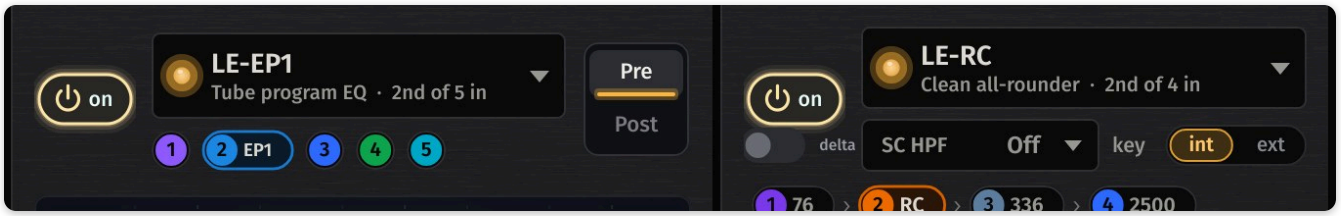
Move the middle five. Drag CHAR EQ, LACHATZ, LACHASH, TONE or TAPE along the strip to change where it sits. The clean-up stages (1964 PK, Reso, Tahor) always come first and the limiter always comes last, so the chain stays safe. A **reset order** pill appears; click it to go back to the standard order. A plain click on TONE or TAPE still opens the drawer.



The strip with Tape dragged to the front of the chain

Pre / post. In the standard order, the pre / post pill swaps the Character EQs and Dynamics: pre = EQ into the compressors, post = compressors first, then EQ.

Inside a module: the chain lanes. Under the Character EQ menu and the Lachatz menu, a row of numbered chips shows the units that are switched in, in the order they run: 1 runs first. Click a chip to edit that unit; drag it left or right to change the order. The menu above says the same thing in words (for example *2nd of 4 in*, or *switched out*), and its list puts the units that are in at the top, numbered. So you can run, say, the LE-336 into the LE-2500.



The chain lanes: LE-M4 runs 1st and LE-EP1 2nd in the Character EQs; LE-76, LE-336 and LE-2500 in Lachatz

Order changes the sound because each stage reacts to what comes before it. These are the moves the new presets are built on:

Chain	What it does	Try it

Tape › EQ › Dynamics	The tape rounds the peaks first, so the compressor works more gently and the result sounds glued, like tracking to tape.	Lead Vocal Tape First, Drum Bus Tape Into 2500, 808 Tape Then Clip
Dynamics › EQ	Compress first, then EQ: boosts aren't pushed back down by the compressor, so air and presence stay open.	Lead Vocal Air After Comp, Master EQ After Glue, Vocal Bus Iron Glue
… › Tape › De-Ess last	The de-esser catches the esses that the EQ, Tone and Tape made brighter.	Lead Vocal Finish, Rap Lead Helios Bite
Two compressors in series	A smooth compressor levels, a punchy one shapes the hits; each works less.	Master Two-Stage Glue, Rap Lead Two-Stage
Dynamics › Tape (hard)	Pumping compression pushed into saturated tape: an effect, not a fix.	Tape Pump Wash, Breakbeat Squash

Top bar

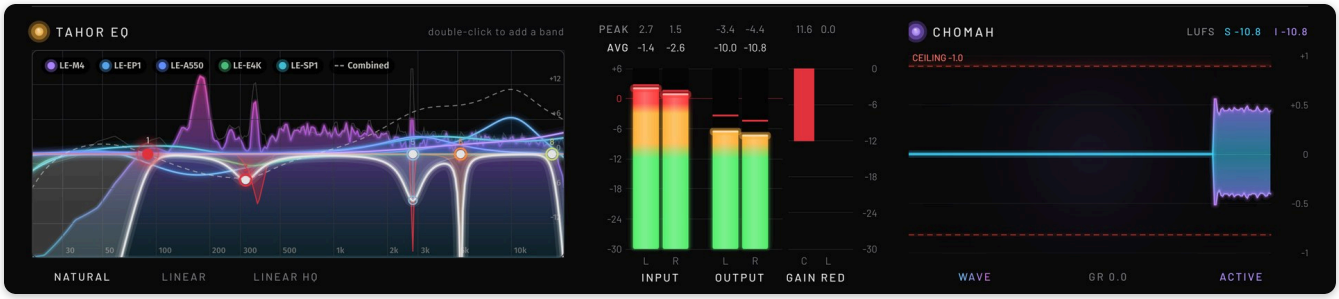


The top bar

Control	What it does
Undo / Redo	Steps back and forward through your changes.
★ ◀ Preset ▶	Click the name to open the preset menu; the arrows step to the previous or next preset. A * after the name means you've changed it. The star marks a favourite.
SAVE / SAVE AS	Saves your own presets. SAVE AS lets you give it a folder.
A > B	Two snapshots of every setting. Click A or B to switch; the arrow copies the current one to the other.
MATCH	When you press BYPASS, the dry signal plays at the same loudness as the processed one, so you judge the sound, not the volume. The reading (for example MATCH -2.4) shows the difference. Your output is never changed.
BYPASS	Bypasses the whole plugin smoothly and stays in time.
OS OFF / 2X / 4X	Oversampling for the whole plugin: cleaner saturation, top-end EQ and limiting, at about 2x / 4x the CPU and a little latency (your DAW compensates). Best changed while stopped.
TRACK	Tracking mode: near-zero latency so a singer can record through the whole chain. Reso pauses and look-ahead stages switch to instant versions. Turn it off for mixing.

⋮ menu	Size (75–125%), Keep Input/Output when changing presets, Reset to Default, Open Presets Folder.

The display



The display: Tahor EQ screen with the Character EQ key, meter bridge and Chomah screen

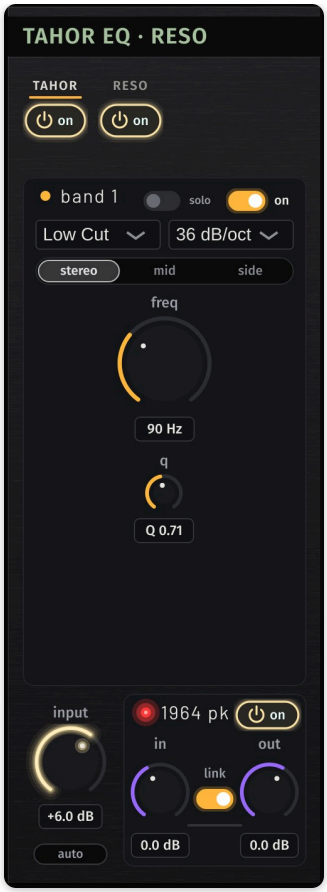
Tahor EQ screen (left). The white curve is the Tahor EQ. Behind it the live spectrum is coloured by level: cyan for quiet, violet in the middle, hot pink at the loudest peaks. The coloured dots are the EQ bands; the red dot is the selected band. Reso's cuts show as red dips hanging from 0 dB, and the Character EQs that are on draw their curves in their own colours, with a dashed Combined curve. Below it: the Natural, Linear and Linear HQ phase modes.

Meter bridge (middle). INPUT and OUTPUT bars (green, yellow at -9 dB, red at -1 dB) with a red peak-hold line, and GAIN RED: C for the compressors, L for the limiter. PEAK and AVG readouts sit on top; click the bridge to reset them.

Limiter screen (right). The wave shows the output level over time, cyan at the centre to violet and hot pink near full scale. A red-orange edge marks where the limiter is working, and the dashed lines are the ceiling. LUFS S is short-term loudness, I is integrated; click the top line to reset.

1 · Tahor EQ, Reso, Input and the 1964 PK

Module 1 cleans the signal before anything adds character: a surgical 8-band EQ, an automatic resonance suppressor, the input level and a peak catcher.



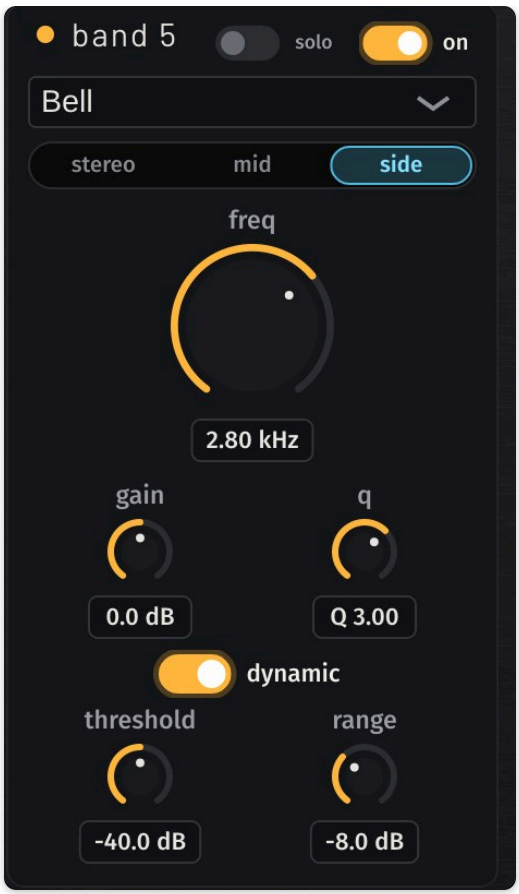
Module 1: Tahor EQ, Reso, Input and the 1964 PK

TAHOR and RESO each have their own on button; click the name above a button to show that tool's controls.

Tahor EQ (8 bands). Double-click the screen to add a band; drag a dot to move it, scroll for width (Q), double-click a dot to remove it, right-click for its type.

- **Types:** Bell, Low Shelf, High Shelf, Low Cut, High Cut, Notch. Cuts have selectable slopes.
- **Band panel:** freq, gain, q, the band's own on switch and solo (hear just that band's region).
- **dynamic:** the band only cuts or boosts when the signal crosses its threshold, by up to its range. Great for harshness that comes and goes.
- **Phase modes:** Natural (zero latency, analog-like), Linear (no phase shift, about 48 ms latency; drums, bass, buses, mastering), Linear HQ (finer low-end resolution, about 176 ms; mixing only).

Stereo placement. On stereo tracks every band can work on stereo (both sides), mid (only the centre: vocal, kick, bass, snare) or side (only the width: reverbs, wide guitars, room). Pick it with the stereo / mid / side pills, or right-click a dot. Mid and Side bands get an M or S tag on the screen, and side bands draw their own dashed cyan curve.



The band panel with the stereo / mid / side pills

- High-pass only the side below about 120–150 Hz to tighten the low end of a mix or master.
- A gentle high shelf on the side adds width and air without making the centre vocal harsher.
- Cut boxiness or harshness in the mid only, and the stereo spread stays untouched.

Reso (resonance suppressor). Finds ringing, honky or harsh frequencies as they happen and turns only those down. It starts at a gentle 15% depth.

Control	What it does
Depth	How much it turns resonances down. Raise from 15% until harshness goes, before the sound dulls.
Sharpness	Narrow, precise cuts (high) or broad, smooth ones (low).
Selectivity	How much a peak must stick out before it's treated.
Attack / Release	How fast it reacts and lets go.
Low / High	The frequency range it works in (drag the edges on the screen too). On vocals, 1.5–9 kHz keeps it off the voice's own harmonics.
Mix	Blend with the untreated signal.

Reso M/S and link. On stereo tracks the pills under Depth choose where Reso listens and works: stereo (one linked detector, the image never moves), mid, side, or m + s (the centre and the width each get their own detector, with mid depth and side depth side by side). In stereo mode, link at 100% treats left and right together; lower it (50–70%) on uneven stereo sources such as double-tracked guitars or a mix bus so a harsh part on one side is tamed without dulling the other.



Reso in m + s: mid depth, side depth and the link knob

Input. Sets the level into the whole strip. Aim for peaks around -12 to -6 dB on the INPUT meter, or use auto.

1964 PK (peak catcher). A fast peak limiter at the front of the chain that catches spikes before the compressors. **in** drives into it (more in = more caught), **out** trims afterwards, and with **link** on, pushing in turns the output down by the same amount, so you get more catching without getting louder. Its light flashes red on each catch.

2 · Character EQs

Eight colour EQs modelled on classic hardware add tone after the clean-up. Use one or stack several; each has its own place in the EQ order and its own m/s setting.



Module 2: Character EQs

EQ	Modelled on	Controls	Sound and best uses
LE-EP1	Tube program EQ (Pultec-style)	Low boost and atten at the same frequency, High boost with width, High atten	Warm low end and a silky top: vocals, bass, buses
LE-E4K	British console EQ	High-pass, Mud, Body, Presence, Air	Forward, punchy mids and presence
LE-N81	Class-A console EQ (1081-style)	High-pass, Mud, Body, Presence, Air	Thick body, rich mids, rock and soul vocals
LE-SP1	Air + silk EQ	High-pass, Presence, Air, Silk	Modern, bright, airy vocals
LE-A550	API 550B-style	Low (30–400 Hz, shelf or peak), Low-Mid (75 Hz–1 kHz), High-Mid (800 Hz–12.5 kHz), High (2.5–20 kHz, shelf or peak), ±12 dB on stepped positions; Filter	Proportional Q (broad for small moves, tighter when pushed) and op-amp punch: drums, bass, rock and rap vocals

LE-K9	SSL 9000K-style	HPF (16–350 Hz) and LPF (3–22 kHz), LF and HF (shelf or BELL), LMF and HMF with Q, ±15 dB; CURVES E or G	Clean, precise console shaping; G curves narrow as you boost or cut. Surgical cuts, mix buses
LE-H69	Helios 69-style	Bass (50 / 60 / 100 / 200 Hz), Mid (700 Hz–6 kHz), Treble (10 kHz), ±10 dB	The most coloured: thick, rounded bass and a brash, forward mid. Drums, guitars, vintage vocals
LE-M4	Maag EQ4-style	Sub, 40, 160, 650, 2.5k (fixed, very broad, ±10 dB) and Air (2.5–40 kHz, up to +15 dB)	Completely clean; the Air band opens the top without harshness. Masters and vocals

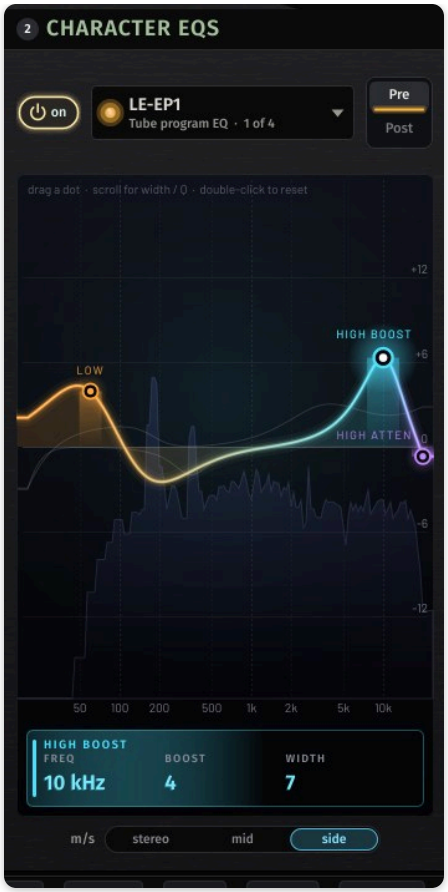


The four console EQs: LE-A550, LE-K9, LE-H69 and LE-M4

Picking, stacking and order. The dropdown shows which EQ you're editing and its place among the ones that are in (for example *2nd of 4 in*). Click its light to switch that EQ in or out. The chain lane under it numbers the EQs that are in, in the order they run; click one to edit it or drag it to reorder. When more are in than fit, the others shrink to numbered dots in their own colours and the one you're editing keeps its name.

The screen. Drag a dot: left/right for frequency, up/down for boost or cut, scroll for width or Q, double-click to reset. The live spectrum behind the curve shows your signal after the EQ, and the strip underneath gives exact values for the selected band.

Mid/side (m/s). The m/s pills under the screen set where the EQ shown works: stereo, mid (lead vocal, kick, bass, snare) or side (reverbs, doubles, wide guitars, room). Each EQ has its own setting, and its tube or console colour follows it, so the other half stays untouched. Stereo tracks only.



The Character EQ with the m/s pills set to side

Tips. LE-A550 high-mid +2 to +4 dB at 3 or 5 kHz gives a snare or a rap vocal bite. LE-H69 bass +3 at 100 Hz and mid +2 at 2.2 kHz is the classic thick, crunchy drum and guitar sound; keep it under +4 on vocals. LE-M4 Air at 20 or 40 kHz, +3 to +6, opens a master or a lead without sibilance, and 160 Hz -2 cleans up mud gently. Air or High Boost on the side only adds width and sparkle without a harsher vocal; a little Body on the mid only firms up the centre.

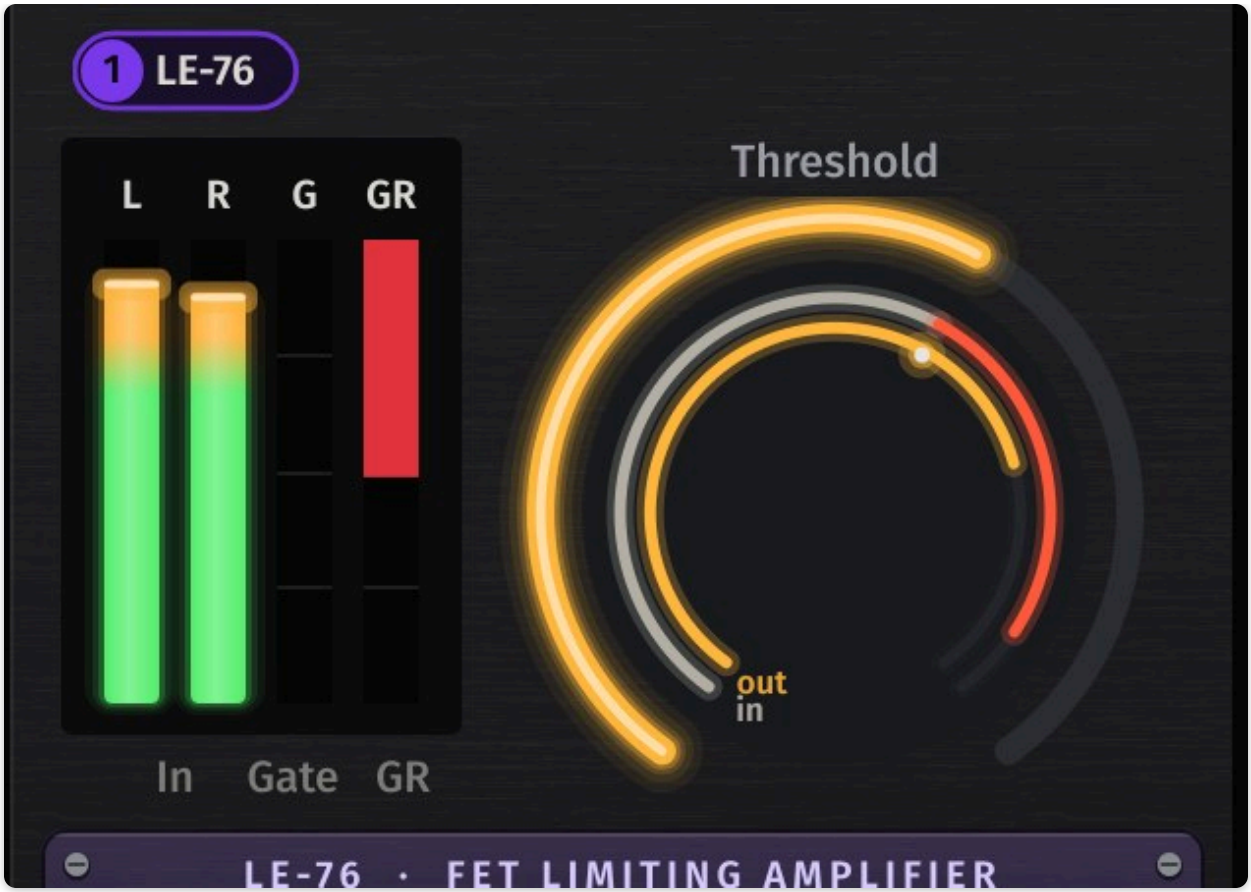
3 · Lachatz — dynamics

Dynamics holds nine compressors and three vocal tools that run before them. Pick a compressor in the dropdown to edit it, click its light to switch it in or out, and use the menu to set the order they run in. You can run several at once; each has its own Mix for parallel compression.



Module 3: Lachatz (dynamics)

The big Threshold knob shows the signal. Inside the gold ring, two live arcs use the threshold's own scale: **in** (grey) is the level going into the compressors and **out** (gold) is what comes out. The part of the input that goes past the threshold turns red: that is what's being squeezed, and the gap between in and out is how much. The knob glows with the signal and brighter while it compresses. (On LE-STA the big knob is Input, so the arcs are hidden.)



The big Threshold knob: the input crossing the threshold in red, the output in gold

Channel compressors



The channel compressors: LE-76, LE-RC, LE-660 and LE-STA

Compressor	Modelled on	Character	Controls

LE-76	1176-style FET limiter	Fast and punchy; grit grows as it compresses	Big Output knob, Attack, Release, Ratio buttons 4 / 8 / 12 / 20, Mix. **Revision:** Purple (modern), Blue Stripe (early, grittier, harder bite), Black (clean and smooth until pushed)
LE-RC	Clean all-round program compressor	Transparent to gently coloured	Ratio, Attack, Release, **Auto Rel**, release shape **Glide** (smooth tail) or **Snap** (livelier), tone **Color** or **Pure**, Make-up, Mix. **Model LE-RV:** a one-knob vocal leveler (3:1, very soft knee) where pulling the threshold down brings the output up automatically
LE-660	Fairchild-style vari-mu	Smooth tube warmth; the ratio rises the harder you push	Threshold, **Time Constant** 1–6 (5 and 6 are program-dependent), Make-up, Mix
LE-STA	Sta-Level-style tube leveler	Thick and velvety, about 3:1, recovers fast then settles slowly	Big **Input** knob (drives into a fixed threshold), Output, Mix, **Mode** Single / Double / Triple, **Recovery** Fast / Medium / Slow / Very Slow

Mastering and bus compressors



The mastering compressors: LE-2500, LE-G, LE-DS1, LE-IRON and LE-336

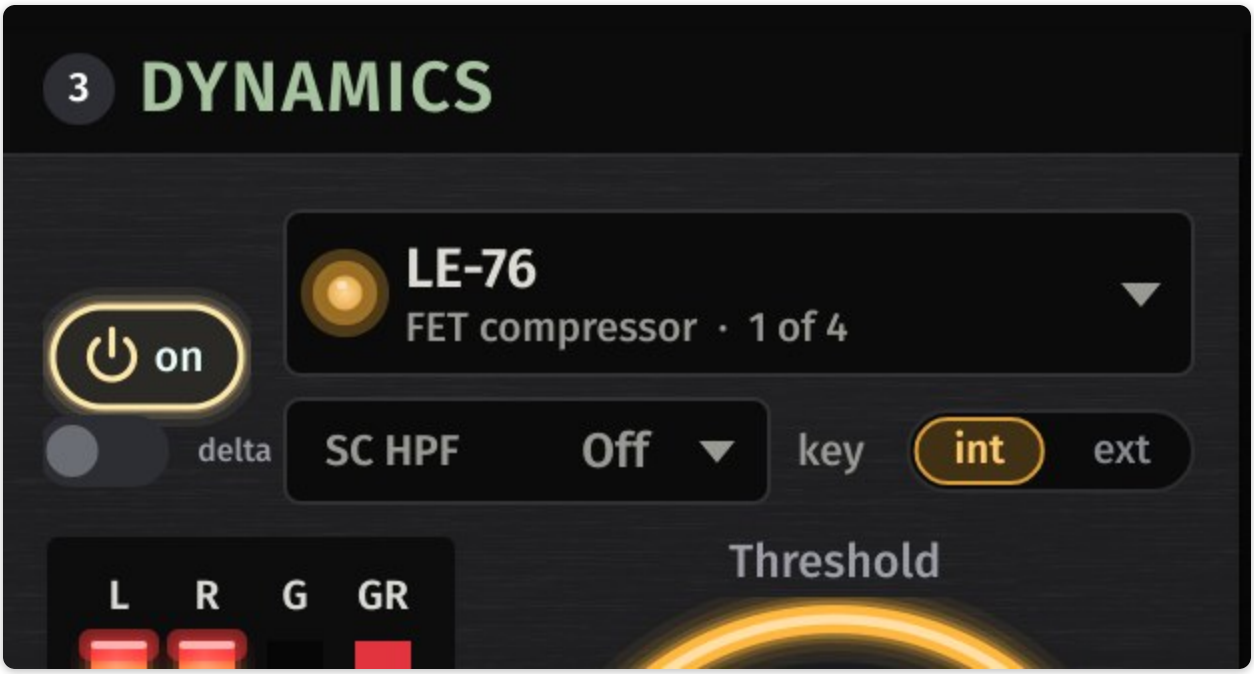
These five are built to take 1 to 3 dB off a mix or a bus and glue it, then let the limiter set the final level. Their switches sit under the knobs; click the arrows to step through them.

Compressor	Modelled on	Character	Controls
LE-2500	API 2500	Punchy and forward; keeps the hits while it holds the mix together	Ratio 1.5–10:1, Attack 0.3 / 1 / 3 / 10 / 30 ms, Release 50 ms–2 s, Make-up, Mix. **Knee** Hard / Med / Soft. **Thrust** Norm / Med / Loud (the detector hears less low end, so the bass doesn't pump the mix). **Type** New (feed-forward, tighter) or Old (feedback, smoother)

LE-G	SSL G-Series bus compressor	The classic console glue; snappy, with a little VCA bite	Ratio 2 / 4 / 10:1, Attack 0.1–30 ms, Release 0.1 / 0.3 / 0.6 / 1.2 s or **Auto**, Make-up, Mix
LE-DS1	Weiss DS1-MK3	Clean and precise, almost no colour	Ratio 1–20:1, Attack 0.1–100 ms, Release 20 ms–3 s, Make-up, Mix. **Knee** Hard / Soft, **Detection** Peak / RMS, **Release** Single or Average (fast and slow together, so long passages settle without pumping)
LE-IRON	SPL Iron	Warm and thick; the ratio grows as you push, tubes add body	Attack 1–6, Release 1–5 or Auto, Make-up, Mix. **Tube Bias** Low to Max (gentle to strong), **Rectifier** Silicon / Germanium / LED (how it reacts), **Airbass** (keeps lows and highs out of the detector)
LE-336	Neve 33609	Rich and solid with a gentle Neve colour, plus a fast limiter	Ratio 1.5–6:1, **Recovery** 100 ms–1.5 s or A1 / A2 (automatic), **Limiter** threshold (Off at 0), Lim Rec, Lim Att Fast / Slow, Make-up, Mix

Glue tips. Start with 2:1, a slow attack (10–30 ms) and Auto release, and aim for 1–2 dB on the GR meter. LE-DS1 when you don't want to hear the compressor; LE-IRON and LE-336 for warmth; LE-2500 for punch; LE-G for the console sound. Turn on SC HPF (or Thrust / Airbass) so the kick and bass don't drive the whole mix.

Shared controls and the vocal tools



The top of Dynamics: delta, SC HPF and the key switch

Control	What it does
Meters	L and R show the input level, G the gate's reduction, GR the total gain reduction of all compressors.
SC HPF	Stops low end (plosives, kick, boom) from triggering the compressors: Off, 80, 120 or 160 Hz.
Key (int / ext)	int = the compressors listen to this track. ext = they listen to your DAW's sidechain input, to duck a beat under a vocal or a bass under the kick. Set the source in your DAW first (Ableton: the Sidechain section of the device; Logic: the Side Chain menu).
Delta	Hear only what the compressors change, level-matched. Switch it off before you bounce.
gate	Closes on quiet noise between phrases. Raise until room noise disappears but words stay intact.
breath	Turns breaths down by up to 24 dB while leaving the singing alone.
rider	An automatic fader that evens out loud and quiet phrases by up to ±12 dB before the compressors, so they work less. It works while Dynamics is on.

4 · Lachash — de-esser

The de-esser tames sibilance (s, sh, t) and other harsh peaks; the GR meter on the right shows how much it's taking.



Module 4: Lachash (de-esser)

Control	What it does
freq	Centre of the sibilance band (2–12 kHz), usually 5–8 kHz for vocals. The slider shows the live spectrum of that range.
threshold	How loud an ess must be before it's reduced. Lower = more de-essing. With auto on it becomes Sensitivity.
range	The most it will turn down, up to 24 dB.
Split / Wide	Split turns down only the harsh band; Wide dips the whole signal briefly on each hit.
listen	Hear only the sibilance band, to find the right freq.
auto	Follows the level of the voice so the threshold doesn't need re-setting for loud and quiet parts.
ext key	Listen to the DAW's sidechain input instead, for example on a reverb or doubles bus keyed from the lead vocal.



De-Ess with listen, auto and ext key

Where to put it. In the standard order the de-esser comes after Dynamics. Drag it to the end of the chain when you're adding brightness with the EQ, Tone or Tape, so it catches the esses they bring up.

5–6 · Tone and Tape

Tone (5) and Tape (6) live in a drawer under the plugin: click TONE or TAPE in the strip at the bottom to open it. Each slot has two models; switch with its model selector.



Modules 5 and 6 in the drawer: Tone (Perach) and Tape (Esh)

Tone — Perach (adaptive tone shaper). Perach listens and keeps steering the tone toward a balanced target, so it adapts as the performance changes.

Control	What it does

amount	How strongly it steers (0–10). Starts at a gentle 2. Above 7 it adds "squash": denser and more finished, but more processed.
Low / Body / Presence / Air graph	Drag each region up for more, down for less (±6 dB of "want"); double-click to reset. 1–2 dB goes a long way.
speed	Slow and smooth (0) to following every syllable (10). Keep vocals around 2–5.
mix	Blend with the dry signal. Starts at 80%. If it sounds over-processed, lower mix first.

Suggested amounts: lead vocals 1–4 at mix 50–80%; backing vocals 4–6; drums and loops 6–8; a mix bus or master 1–3 at mix 30–60%.

Tone — Zerem (style shaper). Pick a style (Pop Vocal, Rap Vocal, Mid Forward, R&B Smooth, Airy Ballad, Podcast, Dense, Drums, Bass, Guitar, Keys, Mix Bus), then set amount and mix.

Tape — Slil (tape machine). Speed 30 / 15 / 7.5 ips (lower = warmer, more head bump), width 1/4", 1/2", 1", 2" (wider = cleaner, bigger low end), drive, mix and optional tape noise. For character without smeared transients, use more drive with the mix around 30–50%. On vocals, 15 or 30 ips keeps the low end tidy; 7.5 ips suits bass, kick and pads.

Tape — Esh (character saturator).

Control	What it does
Character	Clear (natural, fuller), Warm (more low end), Round (warm plus mid glue), Glow (brighter and louder), Ember (loudest; rounds hot highs and sibilance).
process	How hard it works (0–100); the level is roughly made up. 15–40 suits most vocals.
tone	warm / flat / bright tilt.
in / out	Drive into Esh (±10 dB) and trim after it (±6 dB). If the VU sits pinned, back off in first.

M/S drive (stereo tracks). The cyan knob tilts the tape: left (Mid) drives the centre harder to thicken a vocal or a mono-heavy beat, right (Side) drives the sides for width. Level and width stay matched, and the tape's low-end bump stays in the centre.



The tape panel with the M/S drive knob

Watch the stack. Perach, tape and the compressors all add density. If a vocal starts to feel squashed, dial one back; usually tape or Perach amount comes down first. Always level-match with out before you judge.

7 · Chomah — limiter and output

The last module sets the final loudness and keeps peaks under the ceiling; output sets the level leaving the plugin. Three engines sit at the top; the knobs take the engine's colour (violet, teal or orange), while mix (blue), link (pink) and width (lime) keep their own.



Module 7: Chomah with Goliath Classic

Control	What it does
Goliath Classic	One smooth limiter (violet).
Goliath Multi	Tames the low, mid and high bands separately before the final limiter, for louder masters with less pumping (teal).
Goliath XL	A loudness maximizer with nine styles, Character and Transients (orange). See below.
ceiling	The highest peak allowed out (-12 to 0 dB). -1.0 dB is safe for streaming.
threshold	Pushes the signal into the limiter (0 to -24 dB). Lower = louder and more limited.
release / Auto Rel	How fast it recovers (1–500 ms), or adaptive: quick on short peaks, slower under sustained limiting.
upward	Upward compression (0–12 dB) in every engine: lifts the quiet parts toward the loud ones before the limiter. The loudest 6 dB and silence are left alone. Try 2–4 dB on masters.
mix	Blend the limited and unlimited signal.
link	At 100% both sides get the same gain. Lower it (50–80%) on a master when hard-panned hits make the whole mix duck.

soft clip / Clipper	Soft clip adds 2–6 dB of rounded saturation before the limiter; the Clipper pushes 1–6 dB and shaves the tallest peaks just before it. 1–3 dB is the usual range.
True Pk	Also catches the peaks between samples that appear on playback or in MP3 and streaming files, so the ceiling holds there too (the screen says dBTP). Use it on masters; it costs about 1 dB.
width / phase	width: 100% = as is, 0% = mono, up to 200%; everything below about 120 Hz stays in the centre. The phase meter: +1 = mono, +0.3 to +0.8 = healthy stereo, below 0 (red) = parts cancel in mono.
Delta	Hear only what the limiter takes off. If you can hear the music clearly in it, it's working too hard.
output	Final level out of the plugin (-24 to +12 dB).

Loudness target. Click the LUFS readout on the limiter screen and pick -14 (Spotify, YouTube, Apple), -11, -9 or -7 LUFS, then play the loudest part of the song. The threshold moves by itself until the short-term loudness sits on the target, and the screen says *set*. Pick Stop to cancel.

The Goliath XL engine

Click Goliath XL in the engine lights. The XL STYLE menu appears above Soft Clip and Clipper, character replaces release, and transients joins the controls.



Module 7: Chomah with Goliath XL

Style	What it does	Good for
Smart	Adapts its release to the music; slower on bass-heavy material so the low end doesn't distort.	A safe start on anything.
Transient	Very quick recovery after hits.	Drum-heavy mixes, rock, hip-hop.
Loud - Pumping	A long, free release that breathes with the kick.	EDM, when you want the pump.
Loud - Balanced	The all-rounder for loud masters (the default).	Most pop, R&B and rap.
Loud - Crisp	Fast release, bright and a little gritty.	Aggressive, upfront mixes.
Loud - Clip	Shaves up to about 5 dB off the tallest peaks first, then limits. The loudest style.	Maximum level; listen for crunch.
Spectral	Also evens out harsh or boomy frequencies while it pushes.	Dense mixes that get harsh when loud.

Bands	Four bands with their own gain and speed: a kick only pulls down the lows.	Mixes where vocals or pads duck with the kick.
Safe	The gentlest: only catches the peaks and leaves the dynamics alone.	Vocals, acoustic, jazz; catching peaks on a bus.

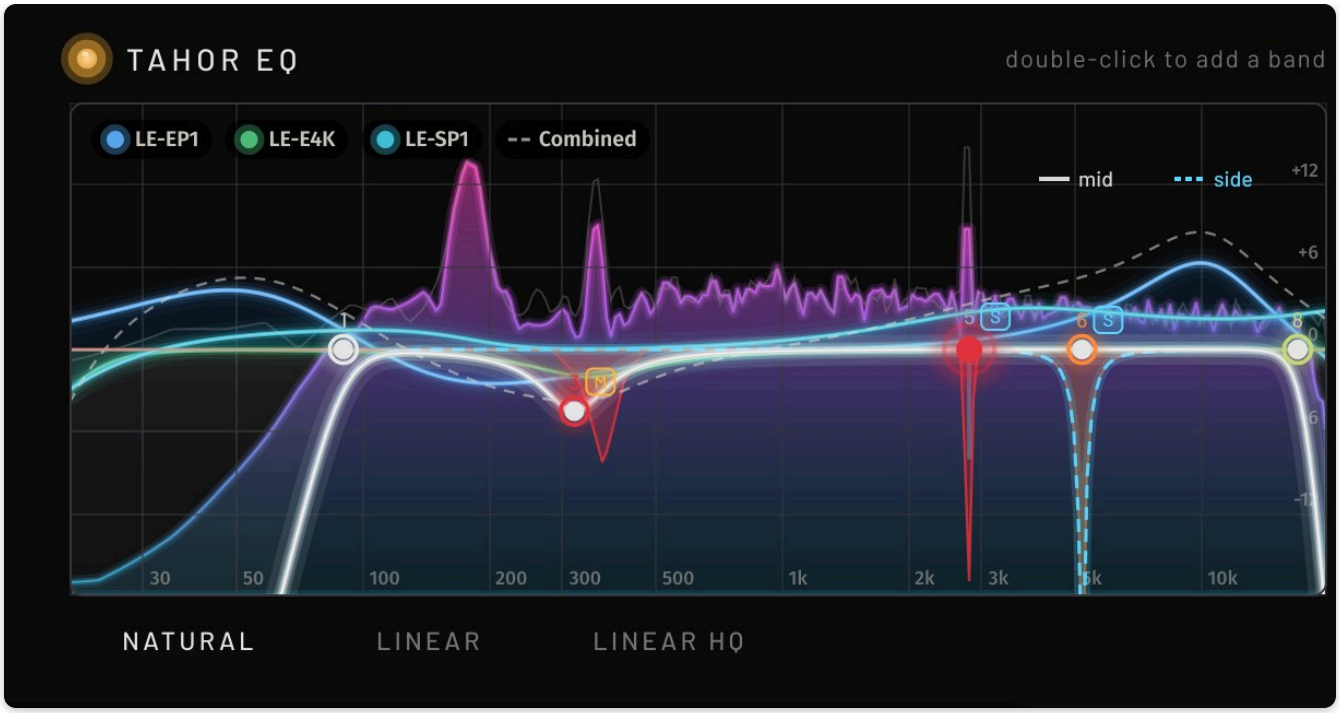
character (0–10, default 5): low = fast, loud, more grit; high = slow and smooth. **transients** (0–100%): lifts the front of each hit before limiting so kicks and snares stay punchy. Start on Loud - Balanced, character 5, transients 0; if the drums lose snap, raise transients to 20–40%; if it pumps, lower character; if it's gritty, raise it.



The limiter: True Pk, delta, width, phase and the clipper

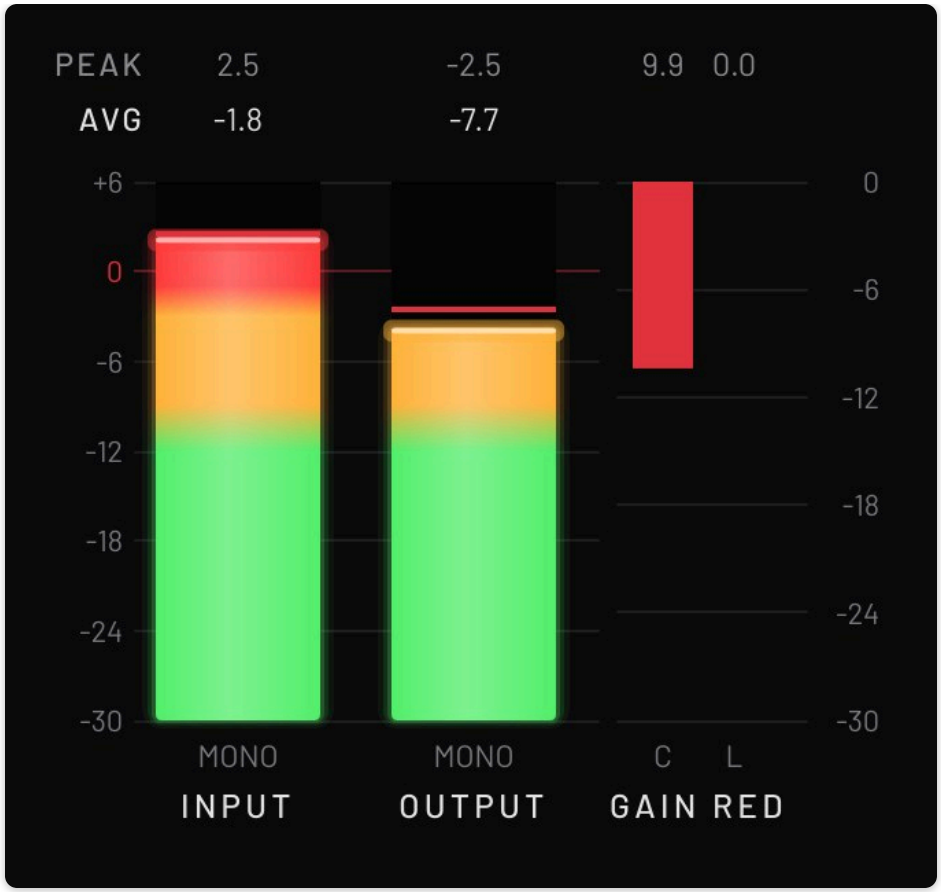
Mid/side and mono tracks

Mid/side works on stereo tracks only and starts switched off everywhere (every band on stereo, Reso on stereo, M/S drive centred). It's available on the Tahor bands, Reso, every Character EQ and the tape's M/S drive.



The Tahor screen with a Mid band, Side bands and the dashed side curve

On a mono track the plugin adapts by itself: the meters show one MONO bar, and the M/S controls, Reso link, limiter link, width and phase are hidden. A preset made on a stereo track still loads: everything counts as mid, so Mid bands still work and Side parts rest.



The meter bridge on a mono track

Keep the low end in the centre: a side high-pass is the safest mid/side move there is. Pushing the sides hard widens a mix but can thin it out in mono, so check in mono now and then.

Presets

The library has 147 factory presets in 11 folders, sorted by what you're working on. The vocal, voice, guitar, keys and FX folders are level-matched on the same real recording, so when you step through them you hear the difference in sound, not volume. The mastering presets are set to their loudness targets instead, and True Peak is on wherever the limiter is.

Folder	Presets	What's in it
Vocals	22	Lead vocals from pop, R&B, soul, rock, country and ballad to the console chains: Lead Vocal Modern, Tape First, Air After Comp, API Forward, SSL Console, Neve Silk and Finish
Stacks & BVs	8	Doubles, stacks, ad-libs, gang shouts; BV Stack Side Air puts the air on the sides only
Rap & Hip-Hop	11	Upfront, dark, trap, drill and doubles; Helios Bite and Two-Stage use the new EQs and chains

Gospel & Worship	10	Power leads, choirs, praise breaks, worship pads; Neve Lift and Tape Air
Podcast & Voice	7	Hosts, guests, voice-over, audiobook, spoken word
Drums	14	Drum buses, live kits, kick, snare, toms, overheads, room
Bass & 808	11	DI, amp, synth bass, 808 weight and harmonics; Bass Iron Weight, 808 Tape Then Clip
Guitars & Keys	13	Acoustic, electric, funk, piano, Rhodes, pads; K9 Bite, Helios Sparkle, Maag Air
Buses	14	Mix bus, music bus, vocal bus and drum bus glue, including SSL, API and Iron chains
Mastering	22	Loud, balanced and streaming masters, one per mastering compressor, the console chains (Neve, API, Weiss) and mid/side masters
FX	14	Telephone, AM radio, lo-fi, megaphone, underwater, saturation, plus Tape Pump Wash, Crushed Parallel Vocal, Distorted Edge Vocal, Small Speaker Mono, Wide Dream Vocal, Breakbeat Squash and Air Blast Whisper

How they were made. Every new preset was run through Goliath on a real recording (a dry male lead, a stereo vocal stack, hip-hop and live drums, an 808, an electric bass, acoustic and electric guitars, a beat and a finished song) and checked for level, true peak and tone. The mastering presets take that song from -13.4 LUFS to between -14 and -8.3 LUFS, all at -1 dBTP. The levels are matched to that material; adjust Input to suit yours.

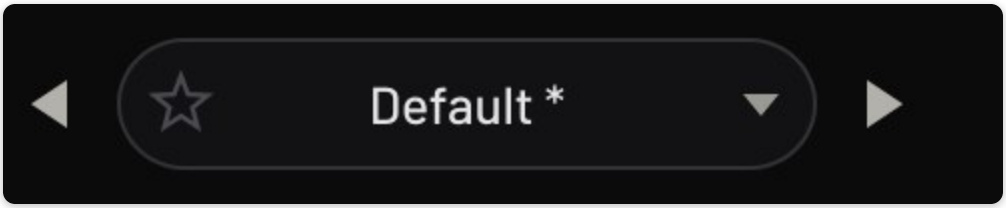
Presets that use the chain order. These show off what moving the modules does; load one and look at the strip at the bottom to see its order.

Preset	Chain	Idea
Lead Vocal Tape First	Tape › EQ › Dynamics › De-Ess › Tone	Tracked-to-tape vocal: LE-N81 and LE-76 after the tape
Lead Vocal Air After Comp	Dynamics › De-Ess › EQ › Tone › Tape	LE-RC levels, then LE-M4 Air opens the top
Lead Vocal Finish	EQ › Dynamics › Tone › Tape › De-Ess	Vocal finishing: LE-M4, LE-DS1, Esh Glow, de-ess last, XL Safe
Rap Lead Helios Bite	EQ › Dynamics › Tone › Tape › De-Ess	LE-H69 mids into a Blue Stripe LE-76, XL Transient
Worship Lead Tape Air	Dynamics › Tape › EQ › De-Ess › Tone	LE-STA into tape, then LE-M4 Air

Drum Bus Tape Into 2500	Tape › EQ › Dynamics	Tape rounds the hits, LE-A550 and LE-2500 glue them
808 Tape Then Clip	Tape › EQ › Dynamics	Harmonics from the tape, LE-2500, then XL Clip
Vocal Bus Iron Glue	Dynamics › EQ › De-Ess	LE-IRON glue, then LE-M4 Air
Master EQ After Glue	Dynamics › EQ › …	LE-G glue first, then LE-M4 tone
Master Two-Stage Glue	LE-336 › LE-2500	Two compressors in series, each doing less
Tape Pump Wash	Tape › Dynamics › EQ	Saturated tape pumped hard by LE-G at 10:1
Distorted Edge Vocal	Tape › EQ › Dynamics	Esh Ember shaped by LE-K9 afterwards
Breakbeat Squash	Dynamics › EQ › Tape	LE-2500 Old at 10:1, LE-H69, tape

Browsing.

- **Favourites:** click the star by the preset name; favourites get their own menu at the top of the list.
- **Search:** Search… at the top of the menu finds presets by name or folder (try *vocal*, *master* or *tape*).
- **Step inside a folder:** tick Arrows step inside this folder, and ◀ ▶ only move through the folder you're in.
- **Keep your levels:** ⋮ › Keep Input/Output when changing presets.
- **Compare:** load one preset into A and another into B and flip between them, with MATCH on.
- **Save your own:** SAVE overwrites your current user preset; SAVE AS asks for a name and a folder. User presets live in their own folder (⋮ › Open Presets Folder) and appear next to the factory ones.



The preset name with its favourite star

Workflows

Mixing a lead vocal.

1. Set Input with auto. Load a Vocals preset close to the style.
2. Tahor: a low cut around 80–100 Hz, a small cut where it's boxy (250–400 Hz). Reso at 15%, range 1.5–9 kHz.

3. Dynamics: rider 6–8 dB to even the phrases, then one compressor taking 3–6 dB (LE-76 for bite, LE-RC or LE-STA for smooth). SC HPF on.
4. Character EQ for presence and air. If boosts get squashed, drag CHAR EQ after LACHATZ.
5. De-Ess at 5–8 kHz; drag it last if Tone or Tape make the esses brighter.
6. Limiter on Goliath XL Safe as a peak catcher, or off if the vocal goes to a bus.

Finishing a vocal (vocal mastering). For a vocal bounce, a podcast or a vocal bus: gentle EQ (LE-M4 Air, a little 160 Hz cut), a clean compressor taking 1–2 dB (LE-DS1, soft knee, RMS, Average release), a touch of Esh Glow, the de-esser last, then Goliath XL Safe with True Pk. That is Lead Vocal Finish.

Mastering a song.

1. Put Goliath on the master. Linear phase on Tahor if you're doing big EQ moves.
2. Tahor: low cut at 20–25 Hz, small moves only; a side high-pass at 120–150 Hz keeps the low end mono.
3. One mastering compressor taking 1–2 dB (see the glue tips), SC HPF at 120 Hz or Thrust / Airbass on.
4. LE-M4 Air or LE-SP1 for the top; Esh Round or Slil at 30 ips for a touch of glue.
5. Limiter: Goliath XL, True Pk on, ceiling -1 dB, then set the threshold or click the LUFS target. Check Delta and the phase meter.

Start from Master Balanced, Master Streaming or one of the compressor masters and adjust.

Buses. On a drum bus try LE-2500 with Thrust Loud, or LE-G at 10:1 with the mix around 45% for parallel smash; on a mix bus LE-G 2:1, 30 ms, Auto; on a vocal bus LE-IRON or LE-RV. Keep the limiter on Safe or off.

Creative effects. Load an FX preset, then experiment with the order: tape before the compressor for pumping saturation, EQ after distortion to shape it, width at 0% and a band-pass for a small speaker.

Tips and troubleshooting

Problem	Fix
macOS won't open the installer	System Settings › Privacy & Security › Open Anyway, or right-click the .pkg and choose Open.
Plugin doesn't show in the DAW	Restart the DAW and rescan (Ableton: Settings › Plug-ins › Rescan). Logic: check it passed the Plug-in Manager scan.
An older session shows the plugin as missing	It was renamed from Mishkal Goliath Strip but kept its internal ID, so a rescan should reconnect it. If it doesn't, tell Lovy.
Singer hears a delay while recording	Turn on TRACK; turn it off again for mixing.

Sound drops for a moment	Changing OS or the Tahor phase mode resets the audio briefly. Change them while stopped.
Too harsh or too dull	Raise Reso Depth a little for harshness; lower it (or narrow Low / High) if it gets dull.
Compressor pumping	Slower release, LE-RC with Glide, LE-STA on Slow, or LE-DS1 with Average release. Turn on SC HPF.
Esses got worse after adding air or tape	Drag LACHASH to the end of the chain.
Too quiet or too loud after a preset	Presets are level-matched to typical material; adjust Input to suit your recording, then Output.
The chain sounds different from the screenshots	Check the strip at the bottom: the preset may use its own order. Click reset order for the standard one.
High CPU	Set OS back to OFF and use Natural phase mode.

Uninstall: delete *Mishkal Goliath Beta.vst3* from /Library/Audio/Plug-Ins/VST3 and *Mishkal Goliath Beta.component* from /Library/Audio/Plug-Ins/Components.

Feedback: tell Lovy what you used it on, which presets you liked, and anything that looked or sounded wrong. This is a beta: don't use it on a mix you can't redo.